

ICASSP 2027 Satellite Workshop on Explainability, Robustness, and Controllability of Language Modeling for Speech and Audio

<https://erc-lm-workshop.github.io>

1 Title

Explainability, Robustness, and Controllability of Language Modeling for Speech and Audio — an ICASSP 2027 Satellite Workshop.

2 Organizers' Information

Full biographies are given in Section 10. Contact details:

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3 Duration / Type

Full-day satellite workshop, to be held before the main ICASSP 2027 conference program.

4 Short Abstract

Large language models, including multimodal variants, have dominated recent machine learning research, and speech and audio are no exception: language-model paradigms now yield impressive results in both understanding and generation. This workshop invites contributions on (i) text-output speech/audio language models (audio-aware LMs), (ii) generative acoustic speech and audio language models, and (iii) audio representation learning underpinning both (e.g., audio tokenizers, self-supervised models). We highlight three cross-cutting themes: *explainability and reasoning*, why a model produces an output and whether its stated reasoning is faithful, including agentic settings where models reason over multiple turns, call tools, and act on a user’s behalf; *robustness*, behavior under noise, adversarial attacks, prompt injection, hallucination, and distribution shift; and *controllability*, steering speaker identity, prosody, rhythm, instrumentation, and instruction-based generation and editing. As audio language models and tokenization become the dominant paradigm, these axes are essential for trustworthy deployment in high-stakes applications such as voice-based assistive and hearing technologies, on-device conversational interfaces, and speaker verification and anti-spoofing. Given the strong momentum and broad interest, we expect a large, interdisciplinary audience spanning the signal processing, speech, audio, and machine learning communities.

5 Expected Attendance

We expect approximately **50 to 100 in-person attendees**. We intentionally favor a relatively compact, discussion-oriented format that encourages interaction between speakers and the audience. Language modeling for speech and audio is one of the most active areas in the field today, sitting squarely within ICASSP’s core speech, audio, and machine-learning tracks; this ensures strong appeal to ICASSP attendees. At the same time, the workshop’s focus on explainability, robustness, and controllability draws participants from adjacent communities, such as trustworthy and explainable machine learning, natural language processing, security, and human–computer interaction. Based on our experience organizing related events (ICASSP 2024 Workshop on Explainable AI for Speech and Audio, Interspeech 2025 Special Session on Explainable Machine Learning for Speech and Audio), we anticipate **40–50 paper submissions**.

6 Publication Intent

Yes. Accepted workshop papers will follow the ICASSP 2027 paper style, format, and length, and will be **published in IEEE Xplore** with the workshop’s separate record number. In accordance with ICASSP policy, an author of each accepted paper must present it in person.

7 Keywords

Audio language models; speech language models; large audio–language models; audio tokenization; self-supervised learning; representation learning; explainability and interpretability; faithfulness; robustness; adversarial attacks; hallucination; controllability; text-to-speech; audio generation; audio editing; trustworthy machine learning.

8 Comprehensive Summary

Motivation. Recent years have witnessed a paradigm shift in machine learning driven by the success of Large Language Models (LLMs). Beginning with the Transformer [1], the field evolved rapidly, with GPT-2 [2] and subsequent variants demonstrating unprecedented capabilities, and catalyzing a broader movement toward multimodal foundation models [3]. Models such as GPT-4V [4], Gemini [5], and LLaVA [6] show that language-model architectures can integrate and reason over multiple modalities, including audio. The speech and audio processing community has been particularly impacted by this shift [7]. Progress has been driven by self-supervised representation learning [8, 9] and discrete tokenization [10], which enable autoregressive language modeling over acoustic signals, and by cross-modal contrastive approaches such as CLAP [11]. Generative audio has advanced substantially through language-model paradigms for speech synthesis [12, 13] and broader audio generation [14, 15, 16, 17, 18, 19]. More recently, agentic approaches have proven a productive direction for these models. First, equipping audio and speech language models with explicit reasoning steps, external tools, and multi-turn planning improves performance on compositional understanding and generation tasks that are difficult to solve in a single forward pass [20, 21]. And recently audio language models have been designed with a dedicated “thinking” mode that generates intermediate reasoning before the final response, improving performance on multi-step audio understanding [22, 23].

Main topics. This workshop is a focal point for research at the intersection of language modeling and speech/audio processing. Our scope centers on acoustic language models, broadly classified into (i) *generative acoustic LMs*, conditioned on preceding audio or text to generate audio [24, 25, 26], and (ii) *audio-aware LMs*, conditioned on audio and/or text to generate text [23, 27, 22] (e.g., for audio understanding). The generative category is further organized by generation approach: diffusion-based methods [16, 28, 29] and language-model-based methods [17, 18, 30]. Alongside these, we (iii) encourage work on *audio representation learning* that supports both, including audio tokenizers and self-supervised models. We organize contributions around three cross-cutting themes:

- **Explainability and reasoning.** Methods and studies that explain *why* an audio/speech language model produces a given output. Examples of such methods include Chain-Of-Thought for LLMs [31, 23], post-hoc attribution methods [32, 33], or models interpretable-by-design [22]. A central concern is *faithfulness*: whether stated reasoning matches what the model computes [34, 35]. This concern sharpens in *agentic* settings, where an audio language model reasons across turns and tool calls: the exposed reasoning trace is then not only an explanation but a control signal for downstream actions, and we welcome work asking whether such traces, and the action sequences they justify, faithfully reflect the underlying computation.
- **Robustness.** How models handle noise, adversarial attacks and prompt injections [34], hallucinations, and temporal shifts, including security, safety, and behavior under distribution shift [36].
- **Controllability.** Architectures and representations that make it easier to control specific parts of generated or edited audio (e.g., speaker identity, prosody, rhythm, instrumentation) and instruction-based generation such as expressive text-to-speech [37, 38] and audio editing [39, 40, 41].

Why a dedicated workshop, and why now? Audio language models and tokenization are among the most dominant approaches in speech and audio today, yet whether we can *understand*,

trust, and steer these models remains comparatively underexplored. Explainability, robustness, and controllability are precisely the properties that decide whether such systems are safe to deploy in sensitive settings such as clinical and assistive speech interfaces and hearing devices, biometric voice authentication, anti-spoofing and audio forensics, and safety-sensitive human-machine voice interaction. We believe that a dedicated venue is warranted as these questions cut across sub-communities that rarely meet in one room, and the pace of progress makes a focused, timely discussion especially valuable.

Complementarity with the main conference and interdisciplinarity. While ICASSP’s main program covers speech and audio methods broadly, it does not provide a concentrated forum on the trustworthiness axes (explainability, robustness, controllability) of audio *language* models specifically. This workshop complements the main conference by convening this emerging topic in a discussion-driven format with invited speakers and a panel. It is inherently interdisciplinary, bridging signal processing, machine learning, interpretability, security, and natural language processing.

Expected impact, outcomes, and dissemination. By bringing together researchers across these areas, we aim to (i) consolidate best practices for evaluating explanations, robustness, and control in audio language models; (ii) surface open problems and shared benchmarks; and (iii) seed collaborations that make the next generation of audio language models more trustworthy. Concrete outcomes include a set of peer-reviewed papers published in IEEE Xplore, recorded/archived invited talks and a panel discussion, and a lasting workshop website (<https://erc-lm-workshop.github.io>) in which we will list the accepted papers. We will disseminate the call and the results widely through ML mailing lists (e.g., ML-News), social media (LinkedIn, X, Bluesky), and institutional Slack channels, building on the audience reached by our prior related events.

9 Planned Components

Contributed papers. We will issue a call for papers on audio language models around the themes above. Submissions will follow the ICASSP 2027 paper style, format, and length, will be double-blind, and will be peer reviewed on OpenReview. We will publicize actively on ML mailing lists (e.g., ML-News), social media (LinkedIn, X, Bluesky), and institutional Slack channels. Aligning with the ICASSP 2027 satellite-workshop timeline, our tentative reviewing schedule is:

- **October 25, 2026** — Workshop paper submission deadline (ICASSP-mandated).
- **November 22, 2026** — Deadline for reviews.
- **December 20, 2026** — Paper acceptance notification (ICASSP-mandated).
- **January 10, 2027** — Final (camera-ready) paper submission (ICASSP-mandated).

We deliberately leave a buffer before the notification date to allow for emergency reviews if needed.

Invited presentations and keynotes. The program includes five invited talks (each ≈ 35 minutes plus 10 minutes for questions), a panel discussion, poster sessions, and oral presentations of contributed papers. Confirmed and tentatively confirmed speakers:

- Prof. Wenwu Wang (University of Surrey), **Confirmed** — audio generative models, language modeling.
<https://www.surrey.ac.uk/people/wenwu-wang>

- Dr. Nick J. Bryan (Adobe Research), **Tentatively Confirmed** — audio GenAI, audio editing in industry.
<https://njb.github.io/index.html>
- Dr. Alexandre Défossez (Gradium), **Confirmed** — audio tokenizers, representation learning.
<https://ai.honu.io/>
- Prof. Dinesh Manocha (University of Maryland), **Confirmed** — audio language models.
<https://www.cs.umd.edu/people/dmanocha>
- Prof. Dilek Hakkani-Tür (UIUC), **Confirmed** — conversational LLM agents.
<https://siebelschool.illinois.edu/about/people/all-faculty/dilek>

All speakers above have been contacted by email and confirmed as indicated. Should any speaker become unavailable, our extensive network in the audio-ML community allows us to substitute quickly.

Panel discussion. The panel will include the invited speakers and be guided around the workshop’s core questions on explainability, robustness, and controllability. We will collect audience questions through a platform such as Slido to involve the audience in the discussion.

Tentative full-day schedule.

1. Opening remarks (9:00–9:15);
2. Plenary speaker 1 (9:15–10:00);
3. Two oral presentations from contributing authors (10:00–10:30);
4. Plenary speaker 2 (10:30–11:15);
5. Poster session and lunch break (11:15–12:45);
6. Plenary speaker 3 (12:45–13:30);
7. Two oral presentations from contributing authors (13:30–14:00);
8. Plenary speaker 4 (14:00–14:45);
9. Coffee break (14:45–15:15);
10. Plenary speaker 5 (15:15–16:00);
11. Poster session (16:00–16:30);
12. Panel discussion (16:30–17:30);
13. Closing remarks (17:30–17:45).

Program committee. Paper evaluation will be shared among the organizers and the following reviewer pool:

- Luca Della Libera, PhD student, Mila + Concordia University — **Confirmed**
- Emanuele Marconato, PhD student, University of Trento — **Confirmed**
- David Beauchemin, PhD student, Université Laval — **Confirmed**
- Steve Azzolin, PhD student, University of Trento — **Tentative**
- Arsenii Gorin, Applied Scientist, Tali AI — **Tentative**
- Zhepei Wang, Research Scientist, Adobe Research — **Confirmed**
- Efthymios Tzinis, Research Scientist, Google — **Tentative**
- Ertug Karamatli, Independent Researcher — **Tentative**
- Benjamin Leblanc, PhD student, Université Laval — **Confirmed**
- Antonio Longa, PhD student, University of Trento — **Tentative**

- Martin Kocour, PhD student, Brno University of Technology — **Confirmed**
- Honza Svec, PhD student, Brno University of Technology — **Tentative**
- Salah Zaiem, Research Scientist, Google DeepMind — **Confirmed**
- Firat Oncel, PhD student, Concordia University — **Tentative**
- Xuechen Liu, PhD student, University of Eastern Finland & Inria — **Confirmed**
- Mirko Bronzi, Senior Research Scientist, Law Zero — **Confirmed**
- Jonah Casebeer, Research Scientist, Adobe / PhD student, UIUC — **Confirmed**
- Krishna Subramani, Research Engineer, Apple — **Confirmed**
- Eleonora Mancini, Research Scientist, Deezer — **Confirmed**
- Ada Tur, PhD student, UIUC — **Tentative**
- Leda Sari, Research Scientist, Otter AI — **Tentative**
- Paris Smaragdis, Full Prof., MIT — **Tentative**
- Pascal Germain, Associate Prof., Université Laval — **Tentative**
- Dimitris Bralios, PhD student, UIUC — **Tentative**
- Jihoon Jeong, PhD student, Mila + Université Laval — **Confirmed**
- Shubham Gupta, PhD student, Mila + Université Laval — **Confirmed**

We have 32 designated reviewers (26 above + 6 organizers). Organizers will each review ≈ 3 papers and share meta-reviewing duties equally. If submissions exceed capacity (max three papers per reviewer), we will recruit additional reviewers from our large networks built through IEEE MLSP 2025 (400+ reviewers), ANNPR 2024/2026, and the ICASSP 2024 XAI-SA workshop; realistically we can reach ≈ 100 reviewers without relying on submitting authors, enabling up to ≈ 100 submissions with at least three reviews each. We will use OpenReview, with which we have extensive experience, and will manage conflicts of interest carefully.

Special requirements and technical needs. Beyond an audio playback system in the room and a projector, we have no special technical requirements.

10 Organizer Biographies

- **Francesco Paissan** — PhD Student at Université Laval and Mila–Québec AI Institute.
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Research Interests: Explainable Machine Learning, Audio Tokenization, Machine Learning for Speech and Audio, On-device Machine Learning.
Prior Organization Experience: Coordinator of the DCASE 2022 “Low-Complexity Acoustic Scene Classification” task; ICASSP 2024 Workshop on Explainable AI for Speech and Audio; Interspeech 2025 Special Session on Explainable Machine Learning for Speech and Audio, CoRL 2026 Workshop on Memory for Robot Foundation Models.
- **Pooneh Mousavi** — PhD Student at Concordia University and Mila–Québec AI Institute.
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Research Interests: Audio Representation Learning, Audio Tokenization, Explainable Machine Learning.
Prior Organization Experience: Conversational AI Reading Group (2024–2026).

- **Aravind Krishnan** — PhD Student at Saarland University.
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Research Interests: Machine Learning for Speech and Audio, Security in Speech Language Modelling, Explainable Machine Learning.
Prior Organization Experience: Interspeech 2025 Special Session on Explainable Machine Learning for Speech and Audio.
- **Dietrich Klakow** – Professor at Saarland University
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Prior Organization Experience: Workshop Beyond Vision and LAnguage: inTEgrating Real-world kNoWledge (LANTERN) (three instances EMNLP 2019, COLING 2020, 2021), Workshop on Weakly Supervised Learning at ICLR 2021, Interspeech 2025 Special Session in Interpretability in Audio and Speech technology.
- **Ricard Marxer** — Full Prof. at Université de Toulon Computer Science Dept. (LIS-CNRS); Visiting Prof. at ÉTS; researcher at ILLS, CNRS and Mila–Québec AI Institute.
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Research Interests: Machine Learning for Speech and Audio, Audio Tokenization, Explainable Machine Learning, Representation Learning.
Prior Organization Experience: Coordinator of Dagstuhl Seminar 2016 on vocal interactivity in-and-between animals, humans and robots (VIHAR); CHiME workshop 2016; VIHAR workshops (2017, 2019, 2021, 2024); special sessions for Interspeech 2016, IEEE IJCNN 2020, EUSIPCO 2024, IEEE IJCNN 2026.
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Prior Organization Experience: NIPS 2018 Workshop on Interpretability and Robustness in Audio, Speech and Language; ICML 2020 Workshop on Self-Supervised Learning for Speech and Audio; Interspeech SpeechBrain Summit 2023; ICASSP 2024 Workshop on Explainable AI for Speech and Audio; General Chair for ANNPR 2024; General Chair for 2025 IEEE Machine Learning for Signal Processing Conference.
- **Cem Subakan** — Assistant Prof. at Université Laval Computer Science and Software Engineering Dept.; Affiliate Assistant Prof. at Concordia University; Associate Academic Member at Mila–Québec AI Institute.
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Prior Organization Experience: Interspeech SpeechBrain Summit 2023; ICASSP 2024 Workshop on Explainable AI for Speech and Audio; Sponsorship Chair for ANNPR 2024; Interspeech 2025 Special Session on Explainable Machine Learning for Speech and Audio; Lead General Chair for 2025 IEEE Machine Learning for Signal Processing Conference.

Additional Information

Concurrent proposals: None of the organizers have recently proposed a similar workshop.

Previous versions: This is the first time we propose this workshop at ICASSP (organizers previously ran the related ICASSP 2024 Workshop on Explainable AI for Speech and Audio and the Interspeech 2025 Special Session).

Diversity statement: The organizing team spans several levels of seniority (two full professors, an associate professor, an assistant professor, two senior PhD students, and a junior PhD student) and multiple geographic regions (Europe and North America). Our invited speakers span institutions, geographic locations (Europe: 2; North America: 3), industry (2) and academia (3), genders, and levels of seniority.

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